

Supplementary Appendices H–K

Full witness construction, computational verification results, applications, and minimal standalone scripts, for

“Typed-Delta Structures: Axioms, Classification, and Verification from 0D to 4D”

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This document is the complete, unabridged version of Appendices H, I, J, and K of the main paper. The main paper itself contains a condensed summary of each appendix with a pointer to this document; nothing here is unique content that contradicts or extends the main paper’s claims — every definition, lemma, check, application, and script below is the full version of something already summarized there. Section numbers below (e.g. “Section 15.8”) refer to sections of the main paper.

Appendix H — A 2D Witness Structure Admitting a Nontrivial Coordinatizing Partition

(Establishes non-vacuity of Theorem 3.6’s relaxed condition by explicit construction; does not establish that every structure satisfying Definition 0.6 admits such a partition.)

H.1 State Space and Operators

Definition H.1 (2D State Space).

Let $X = \{x_0, x_1\}$ and $Y = \{y_0, y_1\}$. Define the state space $S^{(2)} = X \times Y = \{s_{00}, s_{01}, s_{10}, s_{11}\}$, where $s_{ij} = (x_i, y_j)$.

Definition H.2 (Two Coordinate Operators).

$\delta_1(x_i, y_j) = (\neg x_i, y_j)$ (the operator changing only the first coordinate). $\delta_2(x_i, y_j) = (x_i, \neg y_j)$ (the operator changing only the second coordinate). Each acts on exactly one coordinate of $S^{(2)}$ and leaves the other fixed.

Definition H.3 (Operator-Induced Block Systems).

$B_x = \{\{s_{00}, s_{10}\}, \{s_{01}, s_{11}\}\}$ (the partition induced by holding the second coordinate fixed). $B_y = \{\{s_{00}, s_{01}\}, \{s_{10}, s_{11}\}\}$ (the partition induced by holding the first coordinate fixed). Neither block system contains the other.

H.2 Relaxed Differentiation Relation

Definition H.4 (Relaxed 2D Differentiation).

Fix three pairwise distinct values $0, \sigma_1, \sigma_2$. Define $\Delta^{(2)}: S^{(2)} \times S^{(2)} \rightarrow \{0, \sigma_1, \sigma_2\}$ by: $\Delta^{(2)}(s_{ij}, s_{kl}) = 0$ if $(i,j) = (k,l)$; σ_1 if exactly one of $i \neq k, j \neq l$ holds (the pair differs in exactly one coordinate); σ_2 if both $i \neq k$ and $j \neq l$ hold (the pair differs in both coordinates). Equivalently, $\Delta^{(2)}(s,s')$ is determined entirely by the Hamming distance between s and s' as elements of $\{0,1\}^2$: distance 0 maps to 0, distance 1 to σ_1 , distance 2 to σ_2 . That the three values are pairwise distinct is essential to what follows: it is what makes $\Delta^{(2)}$ a genuine relaxation of Axiom 6, which permits only two values, and it is exactly the $n=2$ instance of the typed Hamming-distance relation used for the automorphism-forcing arguments throughout this project.

This differentiation relation distinguishes states at Hamming distance 1 from states at Hamming distance 2 — a structure with no analogue at 1D, where only “different” and “not different” are available. This is precisely the relaxation Theorem 3.6 identifies as necessary, and it instantiates Axiom 8’s two-type requirement exactly: $\Sigma = \{\sigma_1, \sigma_2\}$, with $\Gamma_1(x)$ the three distance-1 neighbors of x and $\Gamma_2(x)$ the single distance-2 neighbor.

H.3 The Automorphism Group of the Relaxed Structure

Lemma H.5 (Automorphisms Preserve Type).

Any bijection $\sigma: S^{(2)} \rightarrow S^{(2)}$ preserving $\Delta^{(2)}$ must send σ_1 -pairs (Hamming distance 1) to σ_1 -pairs, and σ_2 -pairs (Hamming distance 2) to σ_2 -pairs.

Proof.

If σ sent a σ_1 -pair to a σ_2 -pair, it would map a pair with $\Delta^{(2)}$ -value σ_1 to a pair with value σ_2 . Since these values are distinct by Definition H.4, this contradicts σ preserving $\Delta^{(2)}$.

□

Lemma H.6 (The Automorphism Group Has Order Eight and is Dihedral).

$\text{Aut}(S^{(2)}, \Delta^{(2)})$ has order eight, with presentation $\langle r, t \mid r^4 = t^2 = e, trt^{-1} = r^{-1} \rangle$ — the dihedral group of order eight, D_4 .

Proof.

Lower bound (eight explicit automorphisms). Define three bijections of $S^{(2)}$ directly on indices: $\phi_1(s_{ij}) := s_{(1-i)j}$; $\phi_2(s_{ij}) := s_{i(1-j)}$; $\tau(s_{ij}) := s_{ji}$. Each preserves Hamming distance directly — ϕ_1 and ϕ_2 each flip exactly one coordinate, which changes neither pair’s agreement nor disagreement on either coordinate, so the number of coordinates on which any two states differ is unchanged; τ permutes which coordinate is called “first” and which “second,” which likewise does not change how many coordinates two states differ on. Since $\Delta^{(2)}$ depends only on Hamming distance (Definition H.4), all three preserve $\Delta^{(2)}$.

φ_1 and φ_2 each have order two and commute (they flip independent coordinates), generating a subgroup of order four. τ has order two and satisfies $\tau\varphi_1\tau = \varphi_2$ (conjugating by τ swaps which coordinate is flipped, exactly exchanging φ_1 and φ_2); since $\varphi_1 \neq \varphi_2$, $\tau \notin \langle \varphi_1, \varphi_2 \rangle$, so $\langle \varphi_1, \varphi_2, \tau \rangle$ has order eight.

Set $r := \tau\varphi_1$ and $t := \tau$. Since φ_1, φ_2 commute and neither equals the identity, $\varphi_1\varphi_2$ moves every state of $S^{(2)}$, while $(\varphi_1\varphi_2)^2 = \varphi_1^2\varphi_2^2 = e$; and $(\tau\varphi_1)^2 = \tau\varphi_1\tau\varphi_1 = (\tau\varphi_1\tau)\varphi_1 = \varphi_2\varphi_1 = \varphi_1\varphi_2$. Combining these, $(\tau\varphi_1)^2 \neq e$ while $(\tau\varphi_1)^4 = (\varphi_1\varphi_2)^2 = e$, so r has order exactly four. And $trt^{-1} = \tau(\tau\varphi_1)\tau = \varphi_1\tau = (\tau\varphi_1)^{-1} = r^{-1}$, since φ_1 and τ are each their own inverses. This is exactly the dihedral presentation $\langle r, t \mid r^4=t^2=e, trt^{-1}=r^{-1} \rangle$, so $\langle \varphi_1, \varphi_2, \tau \rangle \cong D_4$.

Upper bound (no further automorphisms). Let $g \in \text{Aut}(S^{(2)}, \Delta^{(2)})$. Every state has exactly one σ_2 -partner (Hamming distance 2, obtained by flipping both coordinates) and exactly two σ_1 -partners (Hamming distance 1, obtained by flipping exactly one coordinate); by Lemma H.5, g must send s_{00} 's σ_2 -partner to $g(s_{00})$'s σ_2 -partner, and s_{00} 's two σ_1 -partners to $g(s_{00})$'s two σ_1 -partners (in either order). Hence g is determined by two choices: which of the four states $g(s_{00})$ is, and which of the two remaining states is the image of which of s_{00} 's two σ_1 -partners (a binary choice). So $|\text{Aut}(S^{(2)}, \Delta^{(2)})| \leq 4 \times 2 = 8$.

Combining both bounds, $|\text{Aut}(S^{(2)}, \Delta^{(2)})| = 8$, exhibited explicitly as $\langle \varphi_1, \varphi_2, \tau \rangle \cong D_4$. □

D_4 has order eight, strictly smaller than the order-24 full symmetric group on four elements that Proposition 1.5 would force under Axioms 6–7. The relaxation of Axiom 6 has genuinely loosened the automorphism group, not merely renamed the same structure.

H.4 A Coordinatizing Partition that Survives

Lemma H.7 (Block Systems are $\text{Aut}(S^{(2)}, \Delta^{(2)})$ -Invariant).

B_x and B_y (Definition H.3) are each invariant, as block systems, under the action of $\text{Aut}(S^{(2)}, \Delta^{(2)})$: every automorphism either fixes both block systems setwise or exchanges them with each other, and none splits a block of either system across both.

Proof.

By Lemma H.6, $\text{Aut}(S^{(2)}, \Delta^{(2)}) = \langle \varphi_1, \varphi_2, \tau \rangle$. φ_1 and φ_2 each flip one coordinate while leaving the other fixed, so each fixes both B_x and B_y setwise. Since the subgroup $\langle \varphi_1, \varphi_2 \rangle$ (order four) fixes both block systems setwise, so does every element of it, including $\varphi_1\varphi_2$.

τ exchanges the two coordinates, so it sends each block of B_x (pairs sharing a second coordinate) to a block of B_y (pairs sharing a first coordinate), and vice versa: τ exchanges B_x and B_y as whole block systems. The remaining three elements involving τ (namely $\tau\phi_1$, $\tau\phi_2$, $\tau\phi_1\phi_2$) likewise exchange B_x and B_y .

No element of $\text{Aut}(S^{(2)}, \Delta^{(2)})$ sends a block of B_x to a set that is neither a block of B_x nor a block of B_y — in particular, none sends a pair such as $\{s_{00}, s_{10}\}$ (Hamming distance 1) to a pair such as $\{s_{00}, s_{11}\}$ (Hamming distance 2), since σ_1 -pairs are preserved as σ_1 -pairs by Lemma H.5, and every block of B_x and of B_y is a σ_1 -pair.

□

Proposition H.8 (Existence of a Coordinatizing Partition).

B_x (equivalently B_y) is a coordinatizing partition of $S^{(2)}$ compatible with $\text{Aut}(S^{(2)}, \Delta^{(2)})$ in the sense of Definition 0.7: it is neither the partition into singletons nor the partition consisting of $S^{(2)}$ alone, and by Lemma H.7 it is invariant under the full automorphism group D_4 , not merely under some chosen pair of operators. Equivalently, $S^{(2)} \cong X \times Y$ (Definition H.1) is a nontrivial product decomposition compatible with $\text{Aut}(S^{(2)}, \Delta^{(2)})$, with B_x and B_y the fibers of its two coordinate projections.

The coordinate operators δ_1 and δ_2 (Definition H.2) give this partition a concrete, computable realization: δ_1 acts within each block of B_x and likewise δ_2 within each block of B_y . Both preserve $\Delta^{(2)}$, illustrating concretely how an automorphism can act along one fiber of the partition without disturbing the other. Their presence is illustrative of the partition's structure; it is the partition itself, and its invariance under the full group D_4 established in Lemma H.7, that constitutes the coordinatizing partition required by Definition 0.7. This is the structural feature Theorem 3.5 shows to be entirely absent at 1D, where the forced automorphism group $\text{Sym}(S)$ admits no such invariant partition at any cardinality.

H.5 Relation to Theorem 3.6

Corollary H.9 (Non-Vacuity of the Relaxed Condition).

There exists a structure $(S^{(2)}, \Delta^{(2)})$ satisfying the relaxed condition of Theorem 3.6 (Δ admitting more than the two values Axiom 6 permits) that genuinely admits a coordinatizing partition compatible with its automorphism group (Proposition H.8). Combined with the necessity established in Theorem 3.6, this shows that relaxing Axiom 6 is both a necessary condition for admitting such a partition and a condition satisfiable by at least one concrete structure — not an empty or vacuous escape route.

This corollary establishes existence, not universality: it does not follow, and this paper does not claim, that every structure satisfying Definition 0.6's relaxed condition admits a coordinatizing partition, only that at least one does. Whether the relaxed condition is sufficient in general, for every structure satisfying it, is a separate question left to the full axiomatization of 2D in subsequent work.

Appendix I — Full Computational Verification Results

This appendix gives the complete, unabridged output of the verification package referenced in Section 15.8. Every check constructs its object (witness model, candidate group, distance matrix) directly from definitions and tests the specific named claim; nothing here assumes the claim being tested. All 22 checks passed on every run; total runtime is on the order of 3–4 minutes (the $n \leq 7$ general- n Bose–Mesner sweep and the 3D exhaustive subgroup search are the slow components, each well under a minute individually).

0D (1 check)

- Appendix C, Proposition C.2 (collapse principle) — PASS, illustrative only. Verifies on a finite toy model ($\text{Sym}(n)$ acting transitively on n points, $n=1..6$) that no non-constant function out of the acted-on points is invariant under every permutation — the combinatorial principle the proof's final step invokes. This does NOT construct 0D, which has no internal representation by Axioms 1 and 5; 0D's own claims (Theorem 3.1, Lemma 2.4) remain definitional consequences of the axioms, not independently re-verified here.

1D–2D foundations (5 checks)

- 1D: $\text{Aut}(S, \Delta) = \text{Sym}(S)$ for $|S|=2..6$ — PASS. Every permutation trivially preserves the single-type relation, by definition.
- 2D Lemma 2.7 (C_4/V_4 exclusion at $|S|=4$) — PASS. Verified $|\text{Aut}(S, \Delta)|=8$, transitive, rank=3, matching D_4 's known signature and properly containing any order-4 candidate as a proper subgroup.
- 2D Theorem 3.7 (forced D_4) — PASS. Order=8, transitive=True, rank=3, element-order profile= $\{1:1, 2:5, 4:2\}$, exactly matching D_4 's known profile.
- 2D Theorem 3.8 (Schur-ring dimension 3) — PASS. Linear-independence rank=3, $A_0 + A_1 + A_2 = \text{all-ones}$ confirmed, closure residual=0.00e+00, and the specific witness identity $A_1 \cdot A_1$ decomposes exactly as claimed.
- Axiom 8(iii) chirality necessity — PASS. On an explicit asymmetric-typing counterexample, exactly 4 of D_4 's 8 elements preserve the typing (the rotation subgroup C_4), confirming reflections are excluded by chirality as the axiom requires.

3D, Section 14 (8 checks)

- [Harary 2000] $\text{Aut}(Q_n)=B_n$ at $n=3,4,5$ — PASS. $|B_n|=48, 384, 3840$ respectively, exactly $2^n \cdot n!$ in each case.
- Section 14.3 exhaustive search — PASS (29.4s). Single+pair+triple generator closure over all of B_3 found exactly 3 transitive rank-4 subgroups: B_3 itself (order 48) and two distinct order-24 subgroups.
- E_3/P_3 existence and profiles — PASS. Distinct sets, both order 24, both transitive with rank 4; element-order profiles $E_3=\{1:1,2:9,3:8,4:6\}$ and $P_3=\{1:1,2:7,3:8,6:8\}$, exactly matching the values recorded from the original exhaustive search.
- 3D Lemma 2.8 (Axiom 12 excludes E_3, P_3) — PASS. Induced action sizes on $\Gamma_1(x)$: $B_3 \rightarrow 6, E_3 \rightarrow 3, P_3 \rightarrow 3$ — only B_3 induces the full $\text{Sym}(3)$.
- Theorem 3.9 / Axiom 12' non-vacuity — PASS. On a toy 3-point structure with genuine internal type-splitting, only 2 of 6 permutations preserve the typing, confirming the axiom has real content and is not vacuously satisfied by everything.
- Lemma 2.12 general- n (no internal splitting in $\Gamma_1(x)$) — PASS at $n=3,4,5,6,7$. Internal Hamming distances within $\Gamma_1(x)$ are exactly $\{2\}$ at every n tested.
- 3D Lemma 2.10 (Bose–Mesner dimension 4) — PASS. Rank=4, sum=all-ones confirmed, closure residual=4.44e-16, and the specific witness identity $A_1 \cdot A_1 = 3A_0 + 2A_2$ confirmed exactly.
- 3D Theorem 3.10 (B_3 uniquely satisfies Axiom 12) — PASS. B_3 's induced action on $\Gamma_1(x)$ has size 6, the full $\text{Sym}(3)$.

4D, Section 15 (8 checks)

- E_4/P_4 existence, distinctness, rank — PASS. Both order 192, both transitive, both rank 5, confirmed distinct as sets.
- 4D Lemma 2.11 (Axiom 12 excludes E_4, P_4) — PASS. Induced sizes on $\Gamma_1(x)$: $B_4 \rightarrow 24, E_4 \rightarrow 12, P_4 \rightarrow 12$.
- 4D Theorem 3.15 general- n counting premises — PASS at $n=2,3,4,5$. At each n , $|B_n|=2^n \cdot n!$ exactly, and the induced action on $\Gamma_1(x)$ has size exactly $n!$ (full $\text{Sym}(n)$) in every case.
- 4D Theorem 3.16 (P_n rank $n+1$) — PASS at $n=3,4,5,6$ (6.5s). Orders 24, 192, 1920, 23040 respectively; transitive and rank exactly $n+1$ at every n tested.
- 4D Lemma 2.13 (Bose–Mesner dimension $n+1$) — PASS at $n=2,3,4,5,6,7$ (107.0s). Rank exactly $n+1$ at every n , sum-to-all-ones confirmed at every n , closure residuals all below $5e-15$.
- Signed-axis 8-point structure — PASS. Confirmed faithful (order $384 = |B_4|$) and transitive, but rank exactly 3, not 5 — confirming this smaller B_4 -set does not satisfy Axiom 10 and cannot serve as an alternative witness model.
- AGL(1,32) order-160 refutation — PASS. $992 \bmod 160 = 32$ (nonzero), confirming by Lagrange's theorem alone that no subgroup of order 160 can exist in a group of order 992.

- Reynolds-averaging Bose–Mesner membership — PASS. Across 20 random seeds ($n=4$), maximum membership residual was $1.80e-15$, confirming the averaged matrix lands in the Bose–Mesner algebra to machine precision regardless of input.

Summary: 22 of 22 checks passed; total wall-clock time approximately 150–215 seconds across repeated runs (variation is machine-load noise, not a difference in results — every individual check’s pass/fail status and numeric output is identical across runs, since none of the checks’ outcomes depend on the random seeds used beyond confirming input-independence as designed).

Appendix I.1 — Minimal Standalone Scripts

The full verification package (Appendix I above) is a multi-file Python library, which a reader may reasonably be unwilling to install and run before trusting its results. The six scripts below exist for exactly that situation: each is a single, self-contained file of 14–15 lines of code (excluding the explanatory comment line), using only `itertools` and, where needed, `numpy` — no other dependency, no helper library, nothing to install beyond a standard Python/numpy environment. Each can be read top to bottom in under a minute and checked by hand against the claim its comment states. The exact code and the exact output obtained when run are both given; nothing has been abridged.

Script 1 — B_4 ’s rank (Theorem 3.13/3.15, $n=4$):

```
# Verifies: B4 has rank 5 on  $\{0,1\}^4$  (Theorem 3.13/3.15,  $n=4$  case).
import itertools
S = list(itertools.product([0, 1], repeat=4))
idx = {p: i for i, p in enumerate(S)}
B4 = {tuple(idx[tuple(v[pm[k]] ^ ((f1 >> k) & 1) for k in range(4)]) for v in S)
      for pm in itertools.permutations(range(4)) for f1 in range(16)}
assert len(B4) == 384
seen, orbits = set(), 0
for i in range(16):
    for j in range(16):
        if (i, j) not in seen:
            orbits += 1
            seen |= {(g[i], g[j]) for g in B4}
print(f"|B4|={len(B4)}, rank={orbits}")
assert orbits == 5
```

Output:
|B4|=384, rank=5

Script 2 — E_4 ’s induced action fails to reach $\text{Sym}(4)$ (Lemma 2.11):

```
# Verifies:  $E_4$ ’s stabilizer induces only 12 (not  $24=\text{Sym}(4)$ ) on  $\Gamma_1(x)$ . (Lemma 2.11)
import itertools
S = list(itertools.product([0, 1], repeat=4))
idx = {p: i for i, p in enumerate(S)}
def parity(pm): return sum(pm[i] > pm[j] for i in range(4) for j in range(i+1, 4)) % 2
E4 = {tuple(idx[tuple(v[pm[k]] ^ ((f1 >> k) & 1) for k in range(4)]) for v in S)
      for pm in itertools.permutations(range(4)) for f1 in range(16)
      if (parity(pm) + bin(f1).count("1")) % 2 == 0}
```

```

assert len(E4) == 192
x0 = idx[(0, 0, 0, 0)]
gamma1 = [idx[v] for v in S if sum(v) == 1]
stab = [g for g in E4 if g[x0] == x0]
induced = {tuple(gamma1.index(g[p]) for p in gamma1) for g in stab}
print(f"|stab|={len(stab)}, induced size={len(induced)}")
assert len(induced) == 12

```

Output:
|stab|=12, induced size=12

Script 3 — P_4 's rank matches B_4 's (Theorem 3.16, $n=4$):

```

# Verifies:  $P_4$  (permutation-even subgroup) has rank 5, same as  $B_4$ . (Theorem 3.16,  $n=4$ )
import itertools
S = list(itertools.product([0, 1], repeat=4))
idx = {p: i for i, p in enumerate(S)}
def parity(pm): return sum(pm[i] > pm[j] for i in range(4) for j in range(i+1, 4)) % 2
P4 = {tuple(idx[tuple(v[pm[k]] ^ ((fl >> k) & 1) for k in range(4)]) for v in S)
      for pm in itertools.permutations(range(4)) if parity(pm) == 0 for fl in range(16)}
assert len(P4) == 192
seen, orbits = set(), 0
for i in range(16):
    for j in range(16):
        if (i, j) not in seen:
            orbits += 1
            seen |= {(g[i], g[j]) for g in P4}
print(f"|P4|={len(P4)}, rank={orbits}")
assert orbits == 5

```

Output:
|P4|=192, rank=5

Script 4 — Bose–Mesner algebra dimension (Lemma 2.13, $n=4$):

```

# Verifies: Hamming algebra on  $\{0,1\}^4$  has dimension exactly 5. (Lemma 2.13,  $n=4$ )
import itertools, numpy as np
S = list(itertools.product([0, 1], repeat=4)); N = len(S)
A = [np.zeros((N, N)) for _ in range(5)]
for i, v in enumerate(S):
    for j, w in enumerate(S):
        A[sum(x != y for x, y in zip(v, w))][i, j] = 1
flat = np.stack([a.flatten() for a in A], axis=1)
rank = np.linalg.matrix_rank(flat)
sum_ok = np.allclose(sum(A), np.ones((N, N)))
prod = A[1] @ A[1]
coeffs, *_ = np.linalg.lstsq(flat, prod.flatten(), rcond=None)
resid = np.max(np.abs((flat @ coeffs).reshape(N, N) - prod))
print(f"rank={rank}, sum_ok={sum_ok}, closure_resid={resid:.1e}")
assert rank == 5 and sum_ok and resid < 1e-9

```

Output:
rank=5, sum_ok=True, closure_resid=8.9e-16

Script 5 — the Reynolds-averaging operator's image (a representation-theory fact, not a physics claim; see Section 15.7's discussion of why this does not constitute physical validation):

```

# Verifies: averaging any matrix under  $B_4$  lands exactly in  $\text{span}\{A_0..A_4\}$ . (rep. theory fact)
import itertools, numpy as np
S = list(itertools.product([0, 1], repeat=4)); idx = {p: i for i, p in enumerate(S)}; N = len(S)
B4 = [np.array([1.0 if idx[tuple(v[pm[k]] ^ ((fl >> k) & 1) for k in range(4)]) == j else 0.0

```

```

        for j in range(N)] for v in S]).T for pm in itertools.permutations(range(4)) for fl in
range(16)]
A = [np.zeros((N, N)) for _ in range(5)]
for i, v in enumerate(S):
    for j, w in enumerate(S):
        A[sum(x != y for x, y in zip(v, w))][i, j] = 1
flat = np.stack([a.flatten() for a in A], axis=1)
H = np.random.default_rng(0).standard_normal((N, N)); H = (H + H.T) / 2
H_avg = sum(g @ H @ g.T for g in B4) / len(B4)
coeffs, *_ = np.linalg.lstsq(flat, H_avg.flatten(), rcond=None)
resid = np.max(np.abs((flat @ coeffs).reshape(N, N) - H_avg))
print(f"membership residual = {resid:.1e}")
assert resid < 1e-8

```

Output:
membership residual = 5.0e-16

Script 6 — the corrected anomaly-detection criterion (see Appendix J’s record of the earlier, rejected [1,15]-degeneracy criterion this replaces):

```

# Verifies: corrected anomaly score (Bose-Mesner residual) scores all-ones as
# normal (0), unlike the rejected [1,15]-degeneracy criterion (a rank-1 artifact).
import itertools, numpy as np
S = list(itertools.product([0, 1], repeat=4)); N = len(S)
A = [np.zeros((N, N)) for _ in range(5)]
for i, v in enumerate(S):
    for j, w in enumerate(S):
        A[sum(x != y for x, y in zip(v, w))][i, j] = 1
flat = np.stack([a.flatten() for a in A], axis=1)
def score(M):
    c, *_ = np.linalg.lstsq(flat, M.flatten(), rcond=None)
    return np.max(np.abs((flat @ c).reshape(N, N) - M)) / (np.max(np.abs(M)) or 1.0)
rng = np.random.default_rng(0)
sym = sum(c * a for c, a in zip(rng.standard_normal(5), A))
generic = (lambda r: (r + r.T) / 2)(rng.standard_normal((N, N)))
print(f"symmetric={score(sym):.4f}, all_ones={score(np.ones((N,N))):.4f},
generic={score(generic):.4f}")
assert score(sym) < 1e-6 and score(np.ones((N, N))) < 1e-6

```

Output:
symmetric=0.0000, all_ones=0.0000, generic=1.0802

All six scripts were re-run together immediately before this appendix was finalized; the outputs shown above are exact, not retyped or summarized from an earlier run.

Appendix J — Genuine Applications of the Verified Machinery

This appendix is kept explicitly separate from Appendix I. The previous appendix checks whether **THIS PAPER’S OWN CLAIMS** are correctly implemented; this appendix asks a different question — can the same verified machinery (B_n , Bose–Mesner algebras, Reynolds averaging) solve genuine, falsifiable problems outside the paper? Results here are not evidence for or against the paper’s structural claims, and should not be read as such.

Accepted applications (tested, working, included)

- Symmetry-based denoising: given a signal exactly invariant under a known group and corrupted by genuinely asymmetric noise (not a function of Hamming distance), Reynolds averaging recovers the invariant part. Tested at $n=4$ against a known ground truth across four noise scales (0.5, 2.0, 5.0, 10.0); the recovery improvement factor was stable at approximately $12.47\times$ in every case, independent of noise magnitude, as expected for a linear projection operator. This is explicitly not the same claim as the rejected spin-glass proposal below — denoising recovers a part of the signal the averaging operator is designed to preserve, whereas a ground-state search depends on exactly the information averaging discards.
- Real error-correcting code symmetry diagnosis: applied to the Extended Hamming [8,4,4] code (a standard, currently-relevant code; generator matrix is public-domain mathematical knowledge). The code's weight distribution was confirmed to match the textbook value $\{0:1, 4:14, 8:1\}$, and its real coordinate-permutation automorphism group was computed directly (not assumed) to have order 1344 — a specific fraction ($1/30$) of the full S_8 (order 40320), and far below B_8 . This is a genuine structural fact about a real object: the answer could have come back as any divisor of 40320, or as showing no symmetry at all, and did not have to be 1344.
- Symmetry-deviation anomaly scoring (corrected): scores an input matrix by its Bose–Mesner-algebra membership residual (how far it is from already having B_n symmetry), a continuous and well-defined diagnostic. Tested on five cases with known ground truth: an exactly-symmetric matrix and the degenerate all-ones matrix both scored 0.0000 (correctly identified as having the relevant symmetry); symmetric-plus-tiny-noise scored 0.0339; symmetric-plus-large-noise scored 0.8764; a fully generic random matrix scored 1.1181 — a correct, monotonic ordering from least to most anomalous.

Proposed and rejected (tested and found false, or found untestable)

Recorded here, in the same spirit as Section 15.7's record of rejected mathematical shortcuts, so they are not reintroduced.

- “Reynolds averaging instantly solves NP-hard spin-glass ground-state problems.” Tested directly against 5 independent random Hamiltonians at $n=4$: the ground-state energy of the group-averaged matrix did not match the true ground-state energy in any of the 5 trials, with errors ranging from substantial to off by a factor of several. The reason is structural: a ground state's location depends on the asymmetric details of the specific Hamiltonian, while group-averaging specifically discards exactly that information. This is the logical opposite of the denoising application above, and the two must not be conflated.
- “This proves 4 physical qubits cannot achieve quantum error correction and the system must cross to 5 dimensions.” The numerical conclusion (the relevant bound first holds at $n=5$) is correct, but it restates the independently and

classically known quantum Hamming bound $2^{n-1} \geq 1+3n$ from quantum coding theory, unrelated to anything in this paper's axioms. Presenting it as a finding of this paper's machinery would be a misattribution of a pre-existing result, not a new derivation.

- “Rank-preservation of P_n can instantly find fragile links in a real, partially failed network and plan equivalent-capacity backup routes.” This proposal specifies no actual algorithm connecting the abstract invariant (rank of a group action on an abstract point set) to the concrete task (real, generally asymmetric network topology after node failures). There is no computation to perform here because the claim is not precise enough to be true or false; it is recorded as an example of fluent terminology without a specified mechanism.
- An earlier version of the anomaly-detection idea used the eigenvalue-degeneracy pattern ([1,15] versus [1,1,4,4,6]) as the anomaly signal, framing the all-ones matrix's [1,15] spectrum as evidence of a special “degenerate” or “pre-physical” state. This was checked directly and is wrong: [1,15] occurs for the all-ones matrix purely because that matrix has rank 1 (every row identical), a basic linear-algebra fact with no connection to B_n , Hamming distance, or any physical notion of degeneracy. Any rank-1 symmetric matrix produces a similarly degenerate spectrum, with or without any group symmetry involved. The corrected criterion used above (Bose–Mesner membership residual of the raw input) does not have this flaw, and the accepted anomaly-scoring application above includes the all-ones matrix specifically to demonstrate the correction.

The principle followed throughout this appendix: an accepted application must answer a question that could have come back negative. None of the three accepted applications above work by restating something automatically true of any input (the flaw rejected in Section 15.8's discussion of real-dataset validation) or by noting that two unrelated quantities happen to produce the same number (the flaw in the rejected quantum-bound misattribution above). If a proposed application cannot fail, it does not belong in this record as an accepted result.

Appendix K — Minimal Standalone Scripts (Section 14)

Parallel to Appendix I.1 above, the four scripts below verify Section 14's key claims directly, each a single self-contained file of 11–16 lines (excluding the comment line), using only `itertools` and, where needed, basic arithmetic — no other dependency. Each can be read top to bottom and checked by hand against the claim its comment states. Exact code and exact output are both given.

Script 1 — E_3 , P_3 identity confirmation (Section 14.5.1, the previously-unresolved “second order-24 subgroup”):

```
# Verifies: E3, P3 distinct, order 24, element-order profiles match Section 14.3's search.
import itertools
from collections import Counter
S = list(itertools.product([0, 1], repeat=3)); idx = {p: i for i, p in enumerate(S)}
```

```

def parity(pm): return sum(pm[i] > pm[j] for i in range(3) for j in range(i+1, 3)) % 2
def grp(pred):
    return {tuple(idx[tuple(v[pm[k]] ^ ((fl >> k) & 1) for k in range(3)]) for v in S)
            for pm in itertools.permutations(range(3)) for fl in range(8) if pred(pm, fl)}
E3 = grp(lambda pm, fl: (parity(pm) + bin(fl).count("1")) % 2 == 0)
P3 = grp(lambda pm, fl: parity(pm) == 0)
def order(g):
    cur, k = g, 1
    while cur != tuple(range(8)): cur = tuple(cur[g[i]] for i in range(8)); k += 1
    return k
print(f"|E3|={len(E3)}, |P3|={len(P3)}, distinct={E3 != P3}")
print(f"E3 profile={dict(Counter(order(g) for g in E3))}")
print(f"P3 profile={dict(Counter(order(g) for g in P3))}")

```

Output:

```

|E3|=24, |P3|=24, distinct=True
E3 profile={2: 9, 3: 8, 4: 6, 1: 1}
P3 profile={2: 7, 3: 8, 6: 8, 1: 1}

```

Script 2 — Lemma 2.8's exclusion argument (B_3 induces full $\text{Sym}(3)$; E_3, P_3 do not):

```

# Verifies: B3 induces full Sym(3) on Gamma_1(x); E3, P3 induce only order-3. (Lemma 2.8)
import itertools
S = list(itertools.product([0, 1], repeat=3)); idx = {p: i for i, p in enumerate(S)}
def parity(pm): return sum(pm[i] > pm[j] for i in range(3) for j in range(i+1, 3)) % 2
def grp(pred):
    return {tuple(idx[tuple(v[pm[k]] ^ ((fl >> k) & 1) for k in range(3)]) for v in S)
            for pm in itertools.permutations(range(3)) for fl in range(8) if pred(pm, fl)}
B3 = grp(lambda pm, fl: True)
E3 = grp(lambda pm, fl: (parity(pm) + bin(fl).count("1")) % 2 == 0)
P3 = grp(lambda pm, fl: parity(pm) == 0)
x0 = idx[(0, 0, 0)]
gamma1 = [idx[v] for v in S if sum(v) == 1]
for name, G in [("B3", B3), ("E3", E3), ("P3", P3)]:
    stab = [g for g in G if g[x0] == x0]
    induced = {tuple(gamma1.index(g[p]) for p in gamma1) for g in stab}
    print(f"{name}: |stab|={len(stab)}, induced size={len(induced)}")

```

Output:

```

B3: |stab|=6, induced size=6
E3: |stab|=3, induced size=3
P3: |stab|=3, induced size=3

```

Script 3 — Theorem 3.15's premises hold at $n=3$ (Theorem 3.10 is now unconditional, not solely dependent on the Section 14.3 exhaustive search):

```

# Verifies: Theorem 3.15's counting-argument premises hold at n=3 (B3 forced unconditionally).
import itertools, math
S = list(itertools.product([0, 1], repeat=3)); idx = {p: i for i, p in enumerate(S)}
B3 = {tuple(idx[tuple(v[pm[k]] ^ ((fl >> k) & 1) for k in range(3)]) for v in S)
      for pm in itertools.permutations(range(3)) for fl in range(8)}
x0 = idx[(0, 0, 0)]
gamma1 = [idx[v] for v in S if sum(v) == 1]
stab = [g for g in B3 if g[x0] == x0]
induced = {tuple(gamma1.index(g[p]) for p in gamma1) for g in stab}
print(f"|B3|={len(B3)} (want 2^3*3!={2**3*math.factorial(3)})")
print(f"induced on Gamma_1(x): {len(induced)} (want 3!={math.factorial(3)}, full Sym(3))")
assert len(B3) == 2**3 * math.factorial(3) and len(induced) == math.factorial(3)

```

Output:

```

|B3|=48 (want 2^3*3!=48)
induced on Gamma_1(x): 6 (want 3!=6, full Sym(3))

```

Script 4 — point-set-level sampling check (Section 14.6’s open item): $AGL(3,2)$, a group of order 1344 properly containing B_3 (order 48) acting on the same 8 points, turns out to be 2-transitive and is therefore excluded by Axiom 11 — one of four classically known alternative degree-8 actions tested, none of which gives the rank-4, non-2-transitive combination needed for a genuine counterexample:

```
# Verifies:  $AGL(3,2)$  (order 1344, bigger than  $B_3$ ) on the same 8 points is 2-transitive
# (rank 2), hence excluded by Axiom 11 -- sampling evidence for point-set-level gap.
import itertools
pts = list(itertools.product([0, 1], repeat=3)); idx = {p: i for i, p in enumerate(pts)}
GL32 = [M for M in itertools.product([0, 1], repeat=9)
        if (M[0]*(M[4]*M[8]^M[5]*M[7]) ^ M[1]*(M[3]*M[8]^M[5]*M[6]) ^
            M[2]*(M[3]*M[7]^M[4]*M[6])) % 2]
def mv(M, v): return tuple(sum(M[3*r+c]*v[c] for c in range(3)) % 2 for r in range(3))
AGL32 = {tuple(idx[tuple(a^b for a,b in zip(mv(M, v), t))]) for v in pts}
        for M in GL32 for t in pts}
seen, orbits = set(), 0
for i in range(8):
    for j in range(8):
        if (i, j) not in seen:
            orbits += 1
            seen |= {(g[i], g[j]) for g in AGL32}
print(f"| $AGL(3,2)$ |={len(AGL32)}, rank={orbits}")
assert len(AGL32) == 1344 and orbits == 2
```

Output:
| $AGL(3,2)$ |=1344, rank=2

The other three candidates referenced in Section 14.6 ($PSL(2,7)$ on $PG(1,7)$, order 168, rank 2; the dihedral group D_8 on an 8-cycle, order 16, rank 5; $AGL(1,8)$ on $GF(8)$, order 56, rank 2) were verified the same way and are not reproduced here to keep this appendix to the most illustrative case; none of the four gives a counterexample. All four scripts above, plus these three, were re-run together immediately before this appendix was finalized; the outputs shown are exact, not retyped or summarized from an earlier run.